

OS Unit 1 — Introduction & Process Management

Complete Notes

Operating Systems (OS) · Unit 1 · Introduction & Process Management · 2025

Overview

These notes cover **Introduction & Process Management**. How operating systems manage processes, CPU time, memory, storage and devices — and keep it all safe and concurrent. Each section states the core idea, a worked intuition and the points examiners look for.

1. Operating System Structures

Monolithic, layered, microkernel and modular designs trade performance for isolation.

Why it matters: Operating System Structures appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of operating system structures in one line.
- A small example showing operating system structures in practice.
- Common mistakes students make and how to avoid them.

2. System Calls

System calls are the interface through which user programs request kernel services.

Why it matters: System Calls appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of system calls in one line.
- A small example showing system calls in practice.
- Common mistakes students make and how to avoid them.

3. Process Concept

A process is a program in execution with its own PCB, address space and state.

Why it matters: Process Concept appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of process concept in one line.
- A small example showing process concept in practice.
- Common mistakes students make and how to avoid them.

4. Process States

Processes move between new, ready, running, waiting and terminated states.

Why it matters: Process States appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of process states in one line.
- A small example showing process states in practice.
- Common mistakes students make and how to avoid them.

5. Context Switching

The kernel saves and restores PCB state to switch the CPU between processes.

Why it matters: Context Switching appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of context switching in one line.
- A small example showing context switching in practice.
- Common mistakes students make and how to avoid them.

6. Threads

Threads share a process's address space; user-level and kernel-level threads map via many-to-one, one-to-one or many-to-many models.

Why it matters: Threads appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of threads in one line.
- A small example showing threads in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Operating System Structures · System Calls · Process Concept · Process States · Context Switching · Threads